CodingBN Brunei
Participant Selection Assessment

Before You Start — Please Read This Guide

This assessment is designed to understand how you think and solve problems. It is not only about knowing the right syntax, we want to see your reasoning process. Please answer every question in this document. There are no optional sections.

Document Structure

- Part 1 — Logic & Problem Solving: general logic and case-study questions. No coding knowledge is required. Answer in your own words or with clear step-by-step reasoning.
- Part 2 — Technical: questions on algorithms, databases, and frontend. Everyone must attempt every question, but how you answer depends on your background (see below).

How to Answer Part 2 (Technical)

- If you have NOT learned coding/IT before: you do not need to write real code. You may answer using plain sentences, a flowchart/diagram, or a clear step-by-step explanation of how you would solve the problem.
- If you HAVE learned coding/IT before: you are required to actually build a working product for each applicable question and include the source code. You may submit it either as a link (e.g. GitHub, Google Drive, etc.) written directly in your answer, or as a separate file/folder submitted together with this document, whichever is more convenient for you.
- Answer honestly according to your own experience, there is no need to declare your background separately; this will already be clear from your CV and the way you answer.

Submission

- Write your answers directly below each question in this document (insert text, diagrams, screenshots, or images as needed).
- If you are attaching source code as separate files, please name them clearly (e.g. by question number) and submit them together with this document.
- Make sure every question in Part 1 and Part 2 has an answer before you submit.

Part 1 — Logic & Problem Solving (Mandatory, All Participants)

Answer in your own words or step-by-step reasoning (no code required). What matters is how you think and how clearly you explain your approach to solving the problem.

A. Logic & Patterns

1. You are given the sequence: 2, 4, 8, 16, ... What comes next, and how did you figure out the pattern?

2. There are 8 balls, and one of them is heavier than the rest. You only have a two-pan balance scale (not a digital scale), and you can only use it twice. How would you find the heavier ball?

3. You are asked to sort 10 books from thinnest to thickest, but you can only compare 2 books at a time. Explain, step by step, how you would sort them from start to finish.

4. You save money starting with $1 on Monday, then double the amount every following day ($2 on Tuesday, $4 on Wednesday, and so on). How much would you have saved in total by Sunday? Explain how you worked it out.

5. There are three switches outside a room, and only one light bulb inside the room, connected to just one of the switches. The bulb is off, and you may only enter the room one time to check it. How would you figure out which switch controls the bulb?

6. Look at this pattern: 1, 4, 9, 16, 25, ... What is the next number, and how did you find the pattern?

7. You need to find a specific name in a phone book containing 1,000 names sorted alphabetically. Would you start from page 1 and check every name one by one, or is there a faster way? Explain your approach and why it is faster (or not).

B. Applied Case Studies / Problem Solving

8. Imagine you work at a small grocery store, and every day many items come in and go out. The owner wants to know which items sold the best this week, but the data is still recorded in a plain notebook. If you were asked to help figure this out, how would you start? What information would you need to collect first, and how would you process it?

9. You are asked to build a simple queue system for a small clinic so patients don't need to line up physically. You don't yet know how to build the actual application, but explain: what would the flow look like from a patient arriving to being called in by the doctor? What information would the system need to record?

10. Describe a real experience (not hypothetical) where you were learning something new and hit a wall or a problem you didn't understand at all (in any area, not necessarily IT). How did you eventually find a solution?

11. You need to place 40 students into groups of 4 for a class project, and you want to make sure no group repeats the exact same combination of students from a previous project. How would you approach grouping them fairly?

12. A food delivery app notices that many customers cancel their orders after waiting more than 20 minutes. If you were asked to figure out why this is happening and propose a fix, what steps would you take?

13. You're helping a small library that still tracks borrowed books on paper. They want to know which books are currently overdue. If you had to design a simple way to track this, even without any app, what information would you need to record, and how would you check for overdue books quickly?

14. Two people are arguing about which route is faster to get to the airport, and neither has any proof. How would you determine, in a fair and objective way, which route is actually faster?

Part 2 — Technical (Mandatory, All Participants)

Everyone must answer every question below. If you have not learned coding/IT before, you may answer using plain sentences, a flowchart, or a step-by-step explanation, you do not need to write real code. If you have learned coding/IT before, you are required to build a working product for each applicable question and attach the source code (as a link in your answer, or as a separate file/folder submitted with this document).

B1. Algorithm / Logic Test

1. Given the following input array, write a function to sort it from smallest to largest:

```
Input  : [ 23, 7, 42, 19, 36, 12, 4, 45, 29, 31 ]
Output : [ 4, 7, 12, 19, 23, 29, 31, 36, 42, 45 ]
```

2. Write a function for the Fibonacci sequence, for example:

```
Fibonacci(5)  // returns [0, 1, 1, 2, 3]
Fibonacci(13) // returns [0, 1, 1, 2, 3, 5, 8, 13, 21, 34, ...]
```

3. Write a function to find a sequence of prime numbers based on user input.

```
Input  : 10
Output : 2, 3, 5, 7, 11, 13, 17, 19, 23, 29
```

4. Write a function to find the letter that appears most frequently in a word (case insensitive).
Example: if the given word is 'Hello World', the result should be the letter L.

5. Write a function to count the number of vowels and consonants in a given sentence.

```
Input  : Hello my name is John Doe
Output : 8 Vowels and 12 Consonants
```

B2. Database Test

Student Table

student_id*	student_name	gender	department_id	gpa
1	John Doe	men	1	3.11
2	Jane Doe	women	2	2.8
3	Abdullah	men	1	3.8
4	Aminah	women	3	3.21

Department Table

department_id*	department_name
1	Science
2	Social
3	Politic
4	Tech

1. Write an SQL query to list all students sorted by department name and GPA (ascending). The columns shown must be: department_name, student_name, and gpa.

2. Write an SQL query to list all departments together with the count of students who have a GPA below 3.0.

3. Write an SQL query to list all departments together with the count of students who have a GPA above 3.0.

4. Write an SQL query to list all departments together with their average GPA.

5. Write an SQL query to list all departments sorted by department name, together with the count of male and female students. Example output:

department_name	women	men

B3. Frontend Test

1. Create an HTML page based on the following layout template (Sidebar, Menu, Content, Footer):

Sidebar	Menu
	Content
Footer	

2. Create a simple calculator using HTML and JavaScript (Number 1 input, Number 2 input, a Calculate button, and a displayed Result — e.g. 2 + 3 = 5).

3. Create an HTML form that validates before submitting (e.g. Name, Email, Address fields — a warning message such as 'Please fill out this field' should appear if a required field is left empty).

4. Create an HTML table that can search student names, with columns for Name, Department, and GPA.

5. Create a multistep HTML form (a multi-step registration form with a progress indicator, e.g. a Name field on the first step, followed by additional steps, with a Next button).